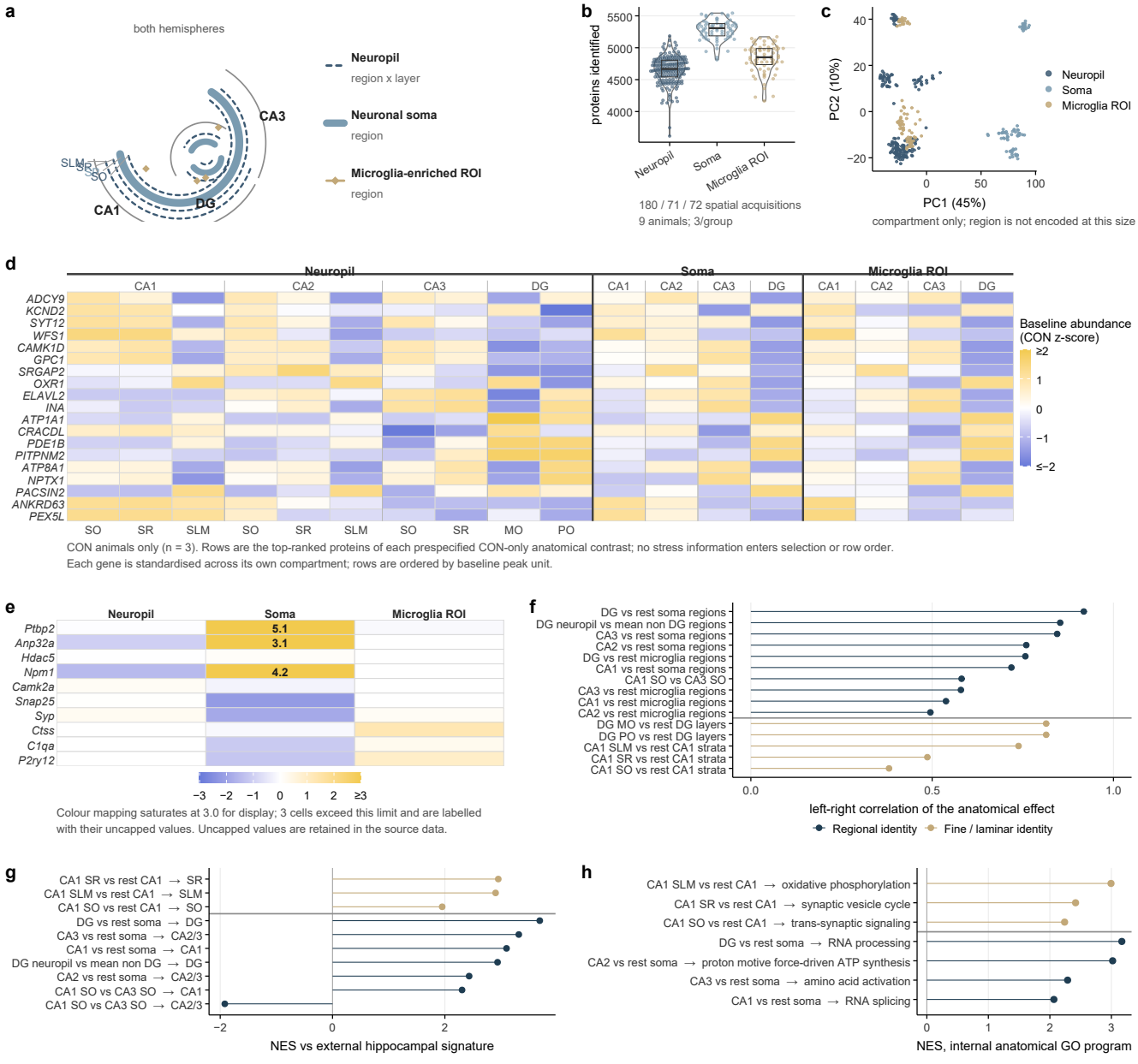

a
both hemispheres
CA3
CA1
DG
SLM
SR
SO
Neuropil
region x layer
Neuronal soma
region
Microglia-enriched ROI
region
b
proteins identified
5500
5000
4500
4000
Neuropil
Soma
Microglia ROI
180 / 71 / 72 spatial acquisitions
9 animals; 3/group
c
PC2 (10%)
PC1 (45%)
40
20
0
−20
0
50
100
Neuropil
Soma
Microglia ROI
compartment only; region is not encoded at this size
d
Neuropil
Soma
Microglia ROI
CA1
CA2
CA3
DG
ADCY9
KCND2
SYT12
WFS1
CAMK1D
GPC1
SRGAP2
OXR1
ELAVL2
INA
ATP1A1
CRACDL
PDE1B
PITPNM2
ATP8A1
NPTX1
PACSIN2
ANKRD63
PEX5L
SO
SR
SLM
SO
SR
SLM
SO
SR
MO
PO
Baseline abundance
(CON z-score)
≥2
1
0
−1
≤−2
CON animals only (n = 3). Rows are the top-ranked proteins of each prespecified CON-only anatomical contrast; no stress information enters selection or row order.
Each gene is standardised across its own compartment; rows are ordered by baseline peak unit.
e
Neuropil
Soma
Microglia ROI
Ptbp2
Anp32a
Hdac5
Npm1
Camk2a
Snap25
Syp
Ctss
C1qa
P2ry12
5.1
3.1
4.2
−3
−2
−1
0
1
2
≥3
Colour mapping saturates at 3.0 for display; 3 cells exceed this limit and are labelled
with their uncapped values. Uncapped values are retained in the source data.
f
DG vs rest soma regions
DG neuropil vs mean non DG regions
CA3 vs rest soma regions
CA2 vs rest soma regions
DG vs rest microglia regions
CA1 vs rest soma regions
CA1 SO vs CA3 SO
CA3 vs rest microglia regions
CA1 vs rest microglia regions
CA2 vs rest microglia regions
DG MO vs rest DG layers
DG PO vs rest DG layers
CA1 SLM vs rest CA1 strata
CA1 SR vs rest CA1 strata
CA1 SO vs rest CA1 strata
0.0
0.5
1.0
left-right correlation of the anatomical effect
Regional identity
Fine / laminar identity
g
CA1 SR vs rest CA1 → SR
CA1 SLM vs rest CA1 → SLM
CA1 SO vs rest CA1 → SO
DG vs rest soma → DG
CA3 vs rest soma → CA2/3
CA1 vs rest soma → CA1
DG neuropil vs mean non DG → DG
CA2 vs rest soma → CA2/3
CA1 SO vs CA3 SO → CA1
CA1 SO vs CA3 SO → CA2/3
−2
0
2
NES vs external hippocampal signature
h
CA1 SLM vs rest CA1 → oxidative phosphorylation
CA1 SR vs rest CA1 → synaptic vesicle cycle
CA1 SO vs rest CA1 → trans-synaptic signaling
DG vs rest soma → RNA processing
CA2 vs rest soma → proton motive force-driven ATP synthesis
CA3 vs rest soma → amino acid activation
CA1 vs rest soma → RNA splicing
0
1
2
3
NES, internal anatomical GO program